# LINFO2315 — Project
# One dimension stabilizer

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## Project Overview

The goal of this project is to design and implement a one-dimensional stabilizer based on an ESP32-S3. The system uses a gyroscope to measure angular deviation from a reference orientation and controls a servo motor to compensate for that deviation.

The servo motor rotation will be proportional to the angle detected by the gyroscope in the reverse direction: small deviations produce small corrections, and larger deviations produce larger corrections. To avoid unnecessary motor motion, a configurable activation threshold will be implemented. To set this threshold, a rotary encoder will be used, allowing the user to adjust the sensitivity of the system in real time.

The threshold needs to be set in real time using the rotary encoder. When the measured angle reaches or exceeds the configured threshold, the servo motor must move according to the proportional control rule. You need to display the current angle and the threshold on a serial monitor for debugging purposes.

The rotary encoder will also be used to control the motor with its push button. A short press will stop the motor from moving, effectively freezing the current servo position. A long press will return the motor to its original (neutral) position.

The final system should continuously read sensor data, apply the threshold logic, and provide reliable, repeatable corrective motion of the servo.

## Project Objectives

The project is divided into the following objectives:

1. **Gyroscope integration:** Interface the gyroscope and compute angular deviation relative to a reference orientation.

2. **Servo motor integration:** Interface the servo motor and attach a visible flag to clearly demonstrate movement.

3. **Proportional correction:** Implement a rule where motor rotation is proportional to the detected gyroscope angle.

4. **Threshold adjustment:** Interface the rotary encoder and use it to set the deviation threshold at runtime.

5. **Conditional actuation:** Move the servo only when the measured deviation reaches or exceeds the configured threshold.

6. **Encoder button behavior:** Use a short press to stop motor updates and a long press to return the servo to its original position.

The accelerometer provided for the project is the MPU6050, which includes both a gyroscope and an accelerometer. You can use the gyroscope to measure angular velocity and compute the angle of deviation. The servo motor should be controlled using PWM signals, and the rotary encoder will provide input for adjusting the threshold and controlling the motor state.

You can find the datasheets for the components here:

- MPU6050:
  `https://www.invensense.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf`
- ESP32-S3:
  `https://www.espressif.com/sites/default/files/documentation/esp32-s3_datasheet_en.pdf`
- Rotary Encoder: on Moodle
- Servo Motor: on Moodle

It is highly recommended to use channels for communication between the different components of your system. This will allow you to decouple the different parts of your system and make it easier to manage the flow of data. Each component (gyroscope reading, threshold adjustment, motor control) can run in its own task, communicating through channels to share data and commands. This will also help you to structure your code and make it more modular.

You need to use ESP-IDF, the fork of FreeRTOS for ESP32, in this project. This will allow you to create multiple tasks for the different components of your system and manage their execution efficiently.

**Warning: Power the gyroscope with 3.3V only to avoid damaging the sensor.**

# Deliverables

1. The project will be written using Rust.
2. You will provide a Readme as documentation for how to use your project.
3. You will also provide a PDF Document including your system design. You'll justify your design choices.
4. You will submit the source code in an archive on moodle.

Deadline for submission: **April 17, 2026 at 18:00**.

There will be a demonstration session, where you will show your project in action. During this session of the demonstration, you will be asked to implement a new feature during one hour followed by the demonstration of this new feature and your project.

If you have any questions or need clarification on any aspect of the project, please don't hesitate to come to Q&A sessions on April 2, or ask questions on the moodle.